Limitations of Self-Submitted Anonymous Graduate Admissions Data

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Paragraph 1 - Selection Bias and Score Inflation

The most fundamental limitation of Grad Café data is self-selection bias: only applicants who actively choose to submit their outcomes are represented in the dataset. This creates a sample that is systematically unrepresentative of the broader applicant population. High-achieving applicants - those accepted to elite programs or those with strong scores to brag about - are disproportionately motivated to share their results. The effect is visible directly in the data: the average GRE Verbal score among reporters in our dataset is 160.75, compared to the ETS-reported national mean of approximately 150-151 for all test takers. Similarly, the average GRE Analytical Writing score of 4.36 exceeds national norms, and the average GPA of 3.77 across all reporters is well above the undergraduate population average. Applicants with unremarkable or disappointing credentials are less likely to submit - and those who are rejected without any standout metric may not report at all - leaving the dataset with an upward-skewed portrait of who actually applies to graduate programs. This gap between the reporting sample and the true applicant pool means that any averages derived from Grad Café data should be interpreted as approximate upper bounds rather than true population means.

Paragraph 2 - Data Reliability and Verification

Beyond selection bias, self-submitted data introduces a separate layer of reliability concerns: there is no verification of the information entered. A user who enters a GRE Analytical Writing score of 99.99 or a GPA of 5.5 faces no correction mechanism; these values entered the raw dataset and had to be filtered out manually. More subtly, applicants may misremember scores, round figures, or conflate different score scales (for instance, reporting a single GRE section score where the field intends a combined score). The university and program name fields are even more susceptible: "Johns Hopkins," "JHU," "John Hopkins" (misspelled), and "Johns Hopkins University" all refer to the same institution, yet appear as distinct entries without normalization. This is precisely why the LLM standardization step from Module 2 was necessary - and its impact was measurable: querying for PhD Computer Science acceptances at top-four universities using raw fields returned zero results, while the same query against LLM-normalized fields returned 28. Taken together, these limitations mean that Grad Café analytics are best used for directional insights and hypothesis generation - identifying rough trends in acceptance rates, comparing programs, or flagging anomalies - rather than as authoritative statistics. Any conclusions drawn should be accompanied by the caveat that the underlying data is voluntary, unverified, and shaped by who chooses to participate.